EEG classifier cut-off sensitivity: replication

The pilot's methods (23 Oct 2026) on the 623 paired patients: GRDA and diffuse slowing from the GRDA-side clinical EEG; N3 from the same patients' PSGs

GRDA and N3: stricter classifier cut-offs select more rhythmic minutes (ACF peak height rises), with large losses in yield. Diffuse slowing (preliminary, 16% of EEG scored) is flat apart from absolute power.

Pattern / classifier	Cut-offs	Retained minutes (ours)	Adults (ours)	Pilot minutes	Pilot adults
GRDA / IIIC.pth	.80 / .90 / .95 / .99	22,512 / 12,303 / 6,151 / 513	357 / 246 / 163 / 40	948 / 349 / 157 / 23	100 / 68 / 44 / 9
Slowing / IIIC <.60	.50 / .60 / .70 / .80	1,415 / 1,337 / 1,242 / 1,108	149 / 141 / 132 / 126	491 / 407 / 372 / 350	92 / 83 / 78 / 75
Slowing / IIIC <.70	.50 / .60 / .70 / .80	1,804 / 1,699 / 1,565 / 1,401	171 / 164 / 152 / 141	738 / 631 / 577 / 538	112 / 105 / 103 / 98
N3 / SLEEPPSG.pth	.50 / .60 / .70 / .80	17,072 / 10,998 / 4,874 / 731	453 / 379 / 246 / 85	233 / 179 / 110 / 37	18 / 18 / 11 / 4
N3 / technologist (reference)	—	39,420	551	archived ss_hm_1: 248 / 237 / 219 / 178	18 / 18 / 18 / 16

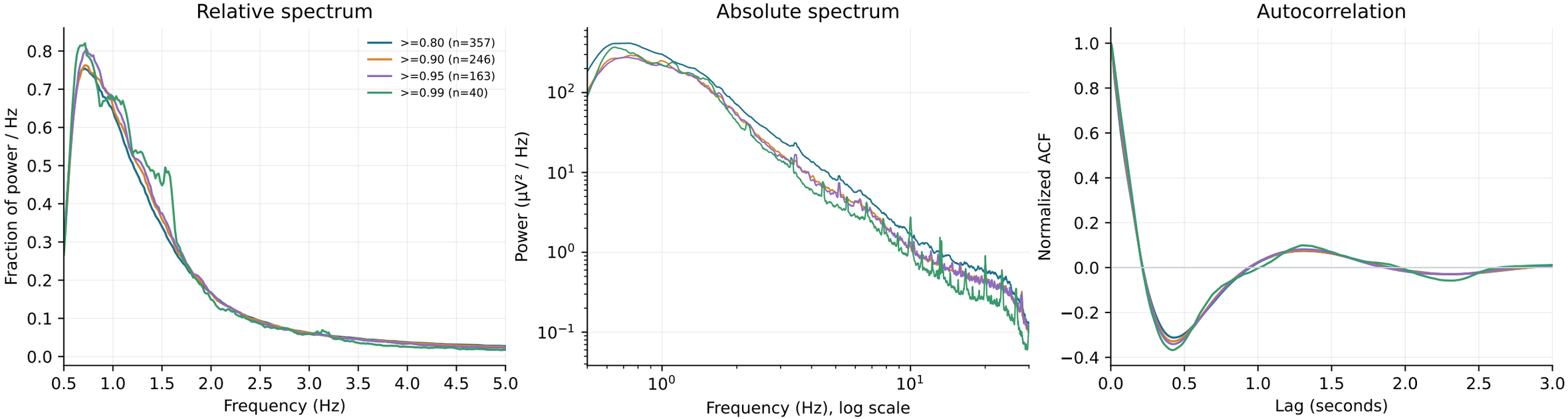
Screened: 623 patients; 790,109 GRDA-side minutes; 766 PSG studies with all six SLEEPPSG derivations (40,846 technologist N3 minutes).

Minutes: whole 60-s blocks; GRDA/slowing 51 contained 1-s predictions (>=41 must pass); N3 11 contained 5-s predictions (>=10). Signals: GRDA side 19-channel common average, 0.5-30 Hz, nine frontal/central electrodes (Fp1 F3 C3 F7 Fz Fp2 F4 C4 F8); N3 F3-M2 F4-M1 C3-M2 C4-M1 (PSG; no common average possible). Thomson multitaper N=12,000, NW=3, K=5; relative power over 0.5-30 Hz; mean-centred ACF normalised at lag 0; first positive peak after the zero crossing at 0.25-2 s. Channels -> minutes per adult -> adults equally. Not reproducible: the archived ss_hm_1 N3 panel (no checkpoint or archived scores for these PSGs); technologist N3 is the reference instead.

GRDA | IIIC.pth

>=41/51 contained 1-s predictions meet each cut-off; GRDA is the highest IIIC class

Curves: all retained adults



Yield at each cut-off; ACF means within the same 40 adults (pilot: 9 adults)

Score >=	Minutes	Adults	ACF height	ACF freq (Hz)	Pilot min	Pilot adults	Pilot height	Pilot freq
>=0.80	22,512	357	0.171	1.039	948	100	0.210	1.117
>=0.90	12,303	246	0.177	1.049	349	68	0.207	1.120
>=0.95	6,151	163	0.186	1.058	157	44	0.209	1.123
>=0.99	513	40	0.200	1.077	23	9	0.228	1.152

Within the same 40 adults

Global cut-off comparison; Holm-adjusted (15 primary tests)

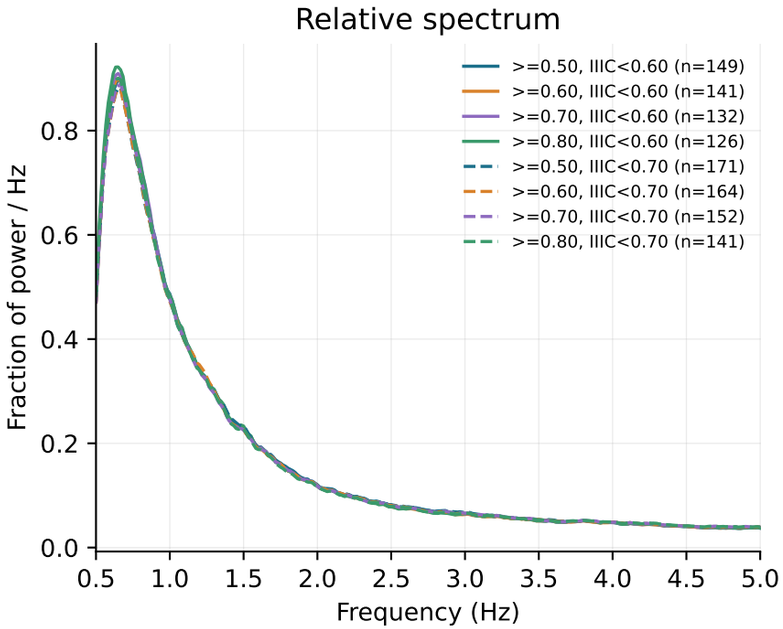
Outcome	Ours	Holm p	Pilot
Relative PSD, 0.5-5 Hz	**	0.00738	ns
Absolute PSD, 0.5-30 Hz	*	0.0333	ns
ACF curve, 0.25-3 s	ns	0.11	ns
ACF peak height	**	0.0046	ns
ACF recurrence frequency	ns	0.63	ns

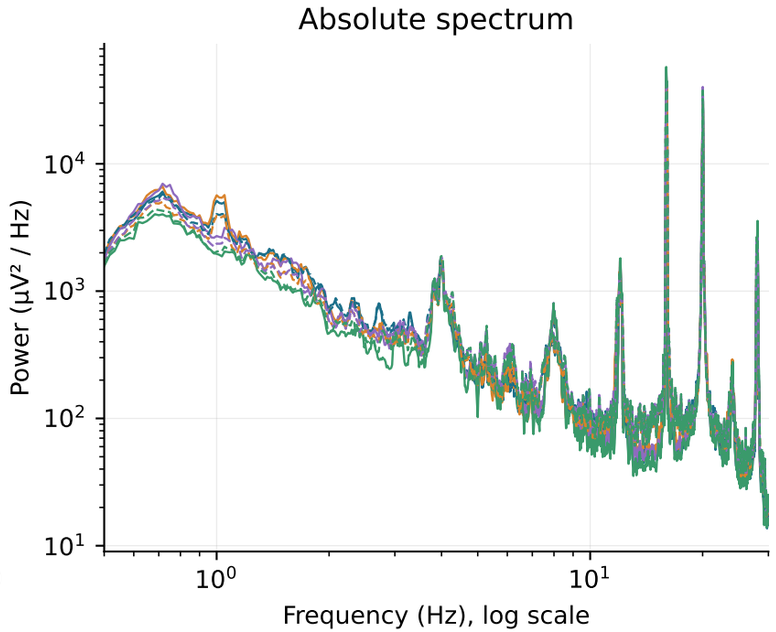
Tests: paired adult sign flips on the differences from the loosest configuration; T = Σ_j (Σ_a d)² / Σ_a d²; exact for n <= 16, else 100,000 flips. * p<.05; ** p<.01; *** p<.001; **** p<.0001; ns p>=.05.

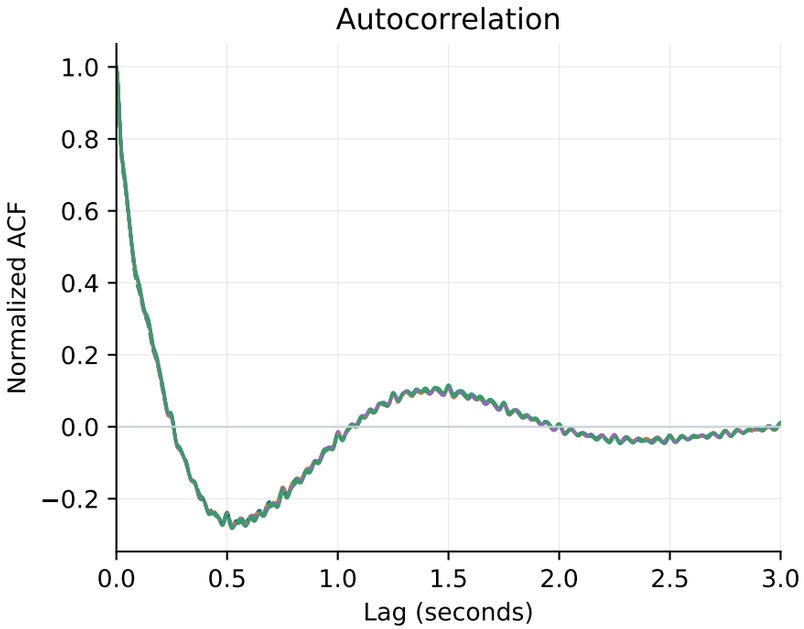
Diffuse slowing | SLOWING.pth

>=41/51 predictions; generalized slowing is the highest class; each IIIC pattern below its ceiling in all 51

Curves: all retained adults







Yield at each cut-off; ACF means within the same 126 adults (pilot: 75 adults)

Score >=	Minutes	Adults	ACF height	ACF freq (Hz)	Pilot min	Pilot adults	Pilot height	Pilot freq
>=0.50, IIIC<0.60	1,415	149	0.141	1.165	491	92	0.122	1.073
>=0.60, IIIC<0.60	1,337	141	0.142	1.165	407	83	0.121	1.072
>=0.70, IIIC<0.60	1,242	132	0.142	1.165	372	78	0.121	1.061
>=0.80, IIIC<0.60	1,108	126	0.142	1.154	350	75	0.121	1.066
>=0.50, IIIC<0.70	1,804	171	0.143	1.172	738	112	0.122	1.077
>=0.60, IIIC<0.70	1,699	164	0.144	1.172	631	105	0.121	1.071
>=0.70, IIIC<0.70	1,565	152	0.144	1.175	577	103	0.122	1.060
>=0.80, IIIC<0.70	1,401	141	0.144	1.163	538	98	0.122	1.052

Within the same 126 adults

Global cut-off comparison; Holm-adjusted (15 primary tests)

Outcome	Ours	Holm p	Pilot
Relative PSD, 0.5-5 Hz	ns	1	ns
Absolute PSD, 0.5-30 Hz	***	0.00015	ns
ACF curve, 0.25-3 s	ns	1	ns
ACF peak height	ns	1	ns
ACF recurrence frequency	ns	1	ns

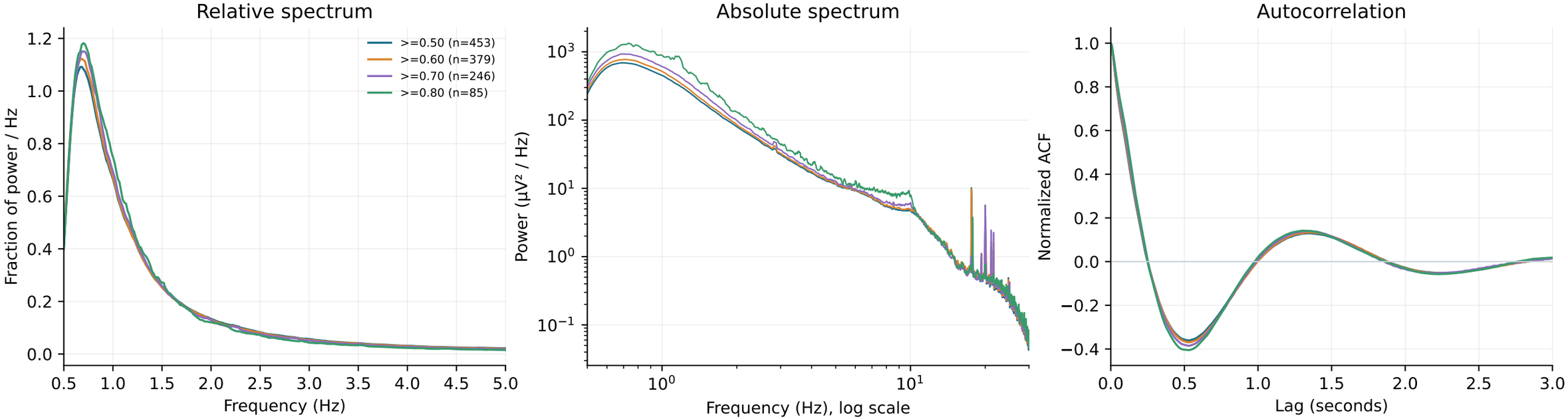
PRELIMINARY: SLOWING.pth had scored 2,994 of 18,661 EEG files (16%) when this was extracted.

Tests: paired adult sign flips on the differences from the loosest configuration; T = Σ_j (Σ_a d)² / Σ_a d²; exact for n <= 16, else 100,000 flips. * p<.05; ** p<.01; *** p<.001; **** p<.0001; ns p>=.05.

N3 | SLEEPPSG.pth (technologist N3 pool)

>=10/11 contained 5-s predictions meet each cut-off; minutes inside technologist-scored N3

Curves: all retained adults



Yield at each cut-off; ACF means within the same 85 adults (pilot: 4 adults)

Score >=	Minutes	Adults	ACF height	ACF freq (Hz)	Pilot min	Pilot adults	Pilot height	Pilot freq
>=0.50	17,072	453	0.178	0.857	233	18	0.174	0.935
>=0.60	10,998	379	0.182	0.854	179	18	0.179	0.934
>=0.70	4,874	246	0.186	0.853	110	11	0.178	0.930
>=0.80	731	85	0.194	0.857	37	4	0.180	0.931

Within the same 85 adults

Global cut-off comparison; Holm-adjusted (15 primary tests)

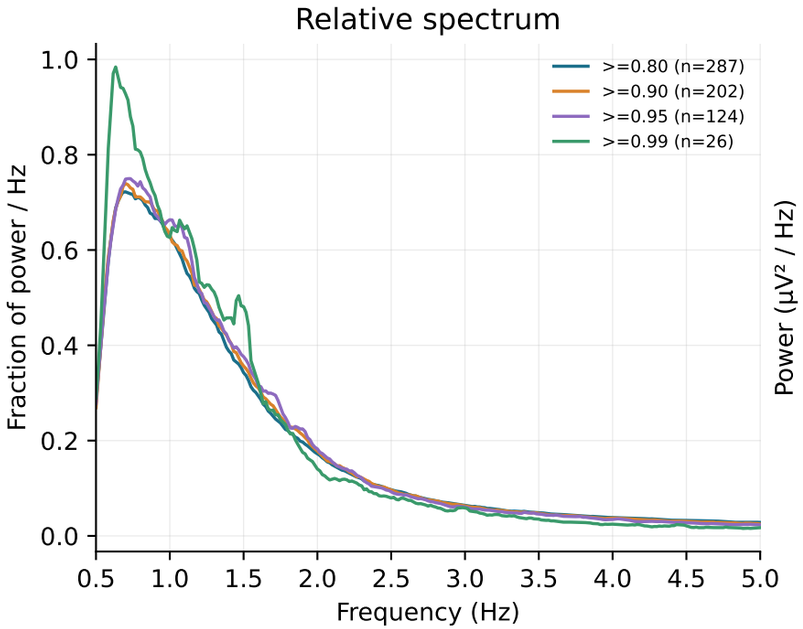
Outcome	Ours	Holm p	Pilot
Relative PSD, 0.5-5 Hz	***	0.00015	ns
Absolute PSD, 0.5-30 Hz	***	0.00015	ns
ACF curve, 0.25-3 s	***	0.00015	ns
ACF peak height	***	0.00015	ns
ACF recurrence frequency	ns	0.749	ns

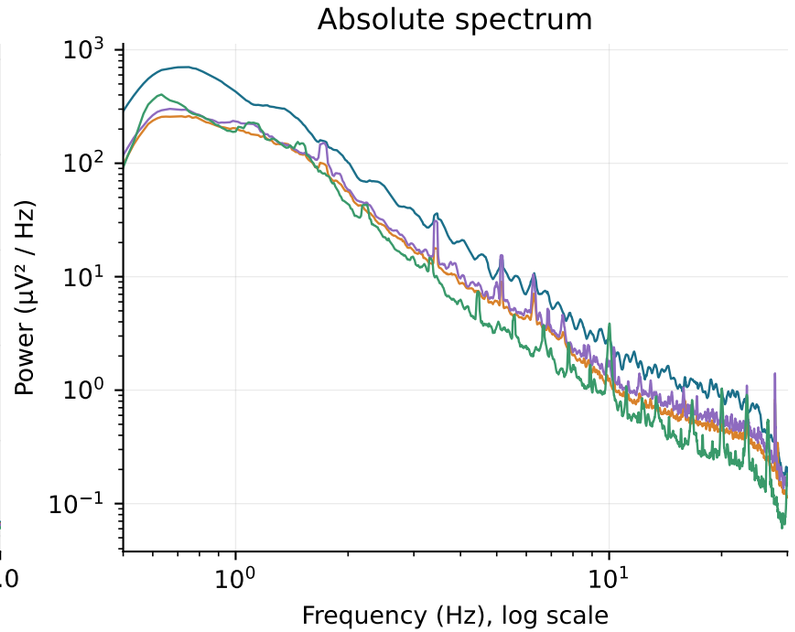
Tests: paired adult sign flips on the differences from the loosest configuration; T = Σ_j (Σ_a d)² / Σ_a d²; exact for n <= 16, else 100,000 flips. * p<.05; ** p<.01; *** p<.001; **** p<.0001; ns p>=.05.

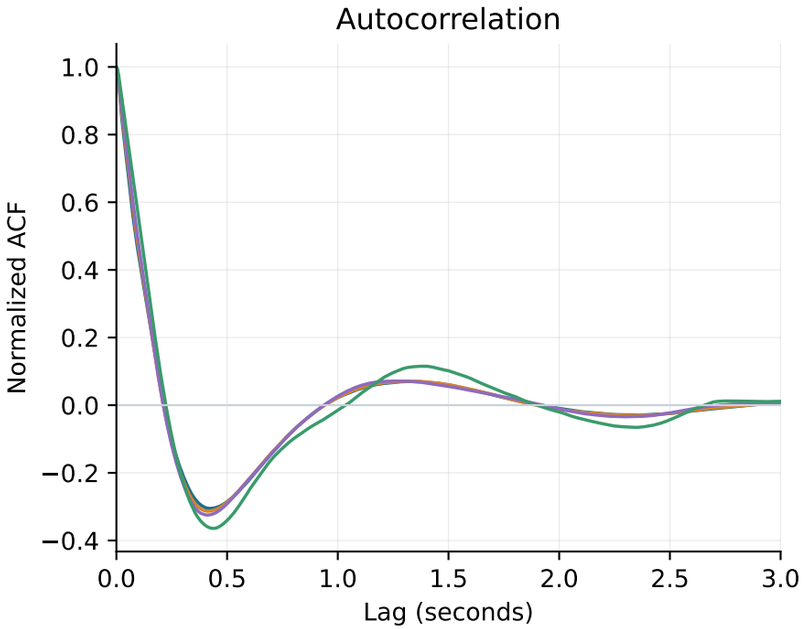
GRDA | IIIC.pth (daytime 08-20, secondary)

>=41/51 contained 1-s predictions meet each cut-off; GRDA is the highest IIIC class

Curves: all retained adults







Yield at each cut-off; ACF means within the same 26 adults

Score >=	Minutes	Adults	ACF height	ACF freq (Hz)
>=0.80	9,612	287	0.183	1.002
>=0.90	5,100	202	0.189	1.011
>=0.95	2,387	124	0.191	1.028
>=0.99	185	26	0.197	1.023

Within the same 26 adults

Global cut-off comparison; Holm-adjusted (daytime family)

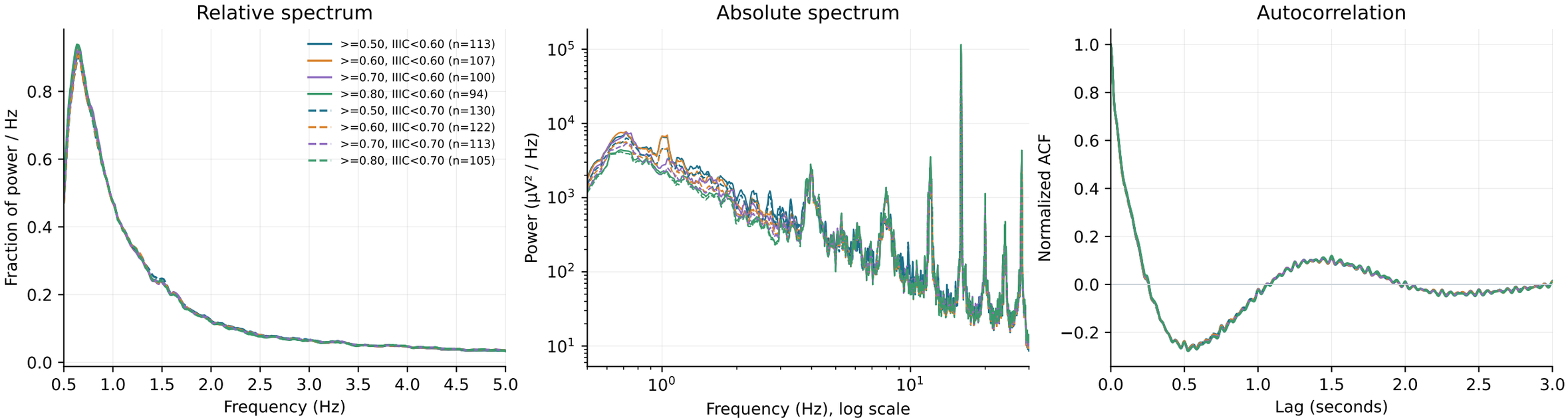
Outcome	Ours	Holm p
Relative PSD, 0.5-5 Hz	ns	0.0505
Absolute PSD, 0.5-30 Hz	ns	1
ACF curve, 0.25-3 s	ns	1
ACF peak height	ns	0.0505
ACF recurrence frequency	ns	0.249

Tests: paired adult sign flips on the differences from the loosest configuration; T = Σ_j (Σ_a d)² / Σ_a d²; exact for n <= 16, else 100,000 flips. * p<.05; ** p<.01; *** p<.001; **** p<.0001; ns p>=.05.

Diffuse slowing | SLOWING.pth (daytime 08-20, secondary)

>=41/51 predictions; generalized slowing is the highest class; each IIIC pattern below its ceiling in all 51

Curves: all retained adults



Yield at each cut-off; ACF means within the same 94 adults

Score >=	Minutes	Adults	ACF height	ACF freq (Hz)
>=0.50, IIIC<0.60	717	113	0.141	1.215
>=0.60, IIIC<0.60	677	107	0.141	1.210
>=0.70, IIIC<0.60	629	100	0.141	1.199
>=0.80, IIIC<0.60	558	94	0.142	1.193
>=0.50, IIIC<0.70	943	130	0.144	1.195
>=0.60, IIIC<0.70	885	122	0.145	1.195
>=0.70, IIIC<0.70	814	113	0.144	1.188
>=0.80, IIIC<0.70	726	105	0.145	1.181

Within the same 94 adults

Global cut-off comparison; Holm-adjusted (daytime family)

Outcome	Ours	Holm p
Relative PSD, 0.5-5 Hz	ns	1
Absolute PSD, 0.5-30 Hz	ns	0.0505
ACF curve, 0.25-3 s	ns	1
ACF peak height	ns	1
ACF recurrence frequency	ns	1

PRELIMINARY: 16% of EEG files scored.

Tests: paired adult sign flips on the differences from the loosest configuration; T = Σ_j (Σ_a d)² / Σ_a d²; exact for n <= 16, else 100,000 flips. * p<.05; ** p<.01; *** p<.001; **** p<.0001; ns p>=.05.